



Department Of Computer Science and Engineering

CSE3422: Software Engineering Lab

Project Report

Project Title: BitBite - Smart Nutrition Platform

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1. Introduction

BitBite is a smart nutrition and health tracking web application designed to help users manage daily food intake, water consumption, body weight, sleep quality, activity-based calorie goals, and personal wellness planning from one platform. The system combines a modern web interface with a PHP backend and a MySQL database so that user accounts, profile data, logs, bookings, article activities, rewards, and admin reports can be stored and retrieved reliably.

The application supports both normal users and an admin role. Users can sign up, complete onboarding, get calorie and macro guidance, log meals, track progress, hire nutritionists or yoga coaches, read Research Hub articles, save articles, complete daily challenges, and claim rewards. The admin side provides platform-level visibility into users, bookings, article activity, rewards, and challenge completion data.

1.1. Motivation

Many people want to improve their health but struggle to maintain a consistent routine because food logging, calorie calculation, water tracking, weight tracking, service booking, and health education are usually scattered across different tools. Students and busy users also need budget-friendly meal guidance that fits local food habits rather than generic diet plans.

A software solution is needed because manual tracking is time-consuming and easy to forget. BitBite addresses this problem by bringing nutrition planning, goal-based recommendations, daily logs, progress visualization, article-based learning, and expert-support booking into one centralized web platform. The system reduces the user effort needed to monitor health habits and also stores activity history in a database for future review.

1.2. Objective

The primary objective of BitBite is to provide a personalized, database-supported nutrition and wellness platform that helps users track food, water, weight, sleep, health goals, expert bookings, and learning activities through a simple web interface.

1.3. Expected Outcome

The expected outcome is a working web application that can run locally using XAMPP, allowing users to create accounts, complete onboarding, manage personal health data, use nutrition tools, book health-related services, interact with Research Hub content, complete challenges, and allow admins to monitor platform activities. The final deliverable includes frontend files, backend PHP logic, and a full MySQL database setup script.

2. Project Description

BitBite is developed as a browser-based health management platform. The application is structured around multiple screens such as login, signup, onboarding, dashboard, meals, food log, AI coach, progress, tools, nutritionists, yoga, research hub, notifications, BMI calculator, calorie calculator, sleep tracker, water tracker, and admin panel.

2.1. List of Features

- User signup, login, logout, and session management
- Password confirmation during account creation
- Six-step onboarding for goal, body information, activity level, diet preference, health condition, and body composition

- Dashboard with health score, calorie summary, macros, water progress, budget usage, wearable-style metrics, charts, and predictions
- Weekly meal plan with budget-based suggestions and shopping list
- Food log add/delete with calories and custom food names saved in the database
- Water tracking from dashboard and full water tracker page
- Weight logging and progress visualization
- Sleep tracking with hours, quality, bedtime, and wake time
- AI Nutrition Coach screen with personalized insight messages
- Health tools: BMI calculator, calorie calculator, macro splitter, steps converter, ideal weight finder, nutrition score
- Nutritionist hiring with service details, payment method, payment status, and transaction information
- Yoga coach hiring with service details, payment method, payment status, and transaction information
- Research Hub articles with read tracking and saved article option
- Rewards Center for target-complete reward claims
- Daily Challenges and streak system for habit building
- Notifications and user settings
- Admin login and admin dashboard for monitoring users, bookings, rewards, challenges, and article activities

2.2. Feature Details

Feature	User Perspective / Working Process
Authentication	The user can create an account with first name, last name, email, password, and confirm password. Login verifies credentials through the backend and opens the dashboard after successful authentication.
Onboarding	After signup, the user completes a six-step setup. The system collects goal, age, gender, height in feet/inches, weight, target weight, body fat, activity level, diet preference, health condition, budget, and wearable preference.
Dashboard	The dashboard summarizes daily calories, macro targets, steps, calories burned, budget spending, water intake, wearable-style statistics, weekly calorie trend, daily streak, and AI prediction information.
Meal Plan	The meal page displays multiple-day meal suggestions with calorie, macro, and cost details. It also includes a shopping list for meal ingredients.
Food Log	Users can add custom foods by entering a food name and calories. Logged food appears immediately on the website and is stored in the food_logs table. Users can also delete food entries.
Water Tracker	Water intake is tracked in cups. Users can update cups from the dashboard or full water tracker, and the database stores the latest cup count for the logged-in user.
Weight Progress	Users can log current weight. The profile weight is updated and weight history is stored for progress tracking.
Sleep Tracker	Users can log sleep hours, quality, bedtime, and wake time. Sleep history supports awareness of recovery and health habits.
AI Coach	The AI Coach screen provides rule-based personalized health insights based on user profile, calorie target, water intake, sleep, and goal.
Nutritionist Hire	Users can browse nutritionist options and submit hire requests with consultation type, fee, payment method, payment status, and notes.
Yoga Coach Hire	Users can browse yoga coaches and submit yoga session requests with session type, fee, payment method, payment status, and notes.
Research Hub	Users can read health-related articles. The system records read activity and also supports saved articles.
Rewards and Challenges	Users can complete daily challenges to build streaks and can claim rewards such as wearable bands or goal-completion rewards.

Feature	User Perspective / Working Process
Admin Panel	Admins can sign in separately and view platform statistics for users, bookings, article reads, saved articles, rewards, and challenge completion.

User Flow

1. A new user opens BitBite and selects Get Started.
2. The user enters signup information and confirms the password.
3. After signup, the user completes onboarding information.
4. The dashboard loads personalized calorie, macro, water, budget, and progress information.
5. The user can log food, water, weight, sleep, and interact with tools or articles.
6. The user can hire a nutritionist/yoga coach, save/read articles, complete challenges, and claim rewards.
7. The admin can log in separately to monitor platform usage and service activities.

3. Technology Used

The BitBite project uses standard web technologies with XAMPP for local hosting. The frontend is built with HTML, CSS, and JavaScript, while the backend uses PHP and MySQL. Chart.js is used for charts and visualization.

3.1. Frontend

Technology	Purpose
HTML5	Structures the website pages, screens, forms, navigation, login/signup views, admin screen, and app sections.
CSS3	Provides responsive styling, cards, navigation, forms, dashboard layout, animations, gradients, and UI theme.
JavaScript	Controls user interactions, API calls, screen routing, validation, calculations, dynamic rendering, chart initialization, and frontend state.
Chart.js	Used to render dashboard charts, calorie rings, macro rings, weekly trends, prediction charts, and nutrition score visualizations.
Google Fonts	Uses Playfair Display and Outfit fonts to create a modern visual identity.

3.2. Backend

Technology	Purpose
PHP	Handles all server-side logic and returns JSON responses to frontend fetch requests.
PDO	Used for MySQL database connection with prepared statements and error handling.
PHP Sessions	Maintains logged-in user session and separate admin session.
Password Hashing	Uses password hashing and verification for secure user/admin authentication.
XAMPP Apache	Runs the PHP application locally on localhost for lab setup and demonstration.

3.3. APIs

The application uses internal REST-like API endpoints through backend.php?action=.... The frontend JavaScript sends GET or POST requests using the Fetch API and receives JSON responses from PHP. No third-party payment API or external AI API is integrated in the current version.

Endpoint / Action	Purpose
signup, login, logout, me	User authentication and session data loading
save_profile	Stores onboarding/profile data
add_food, delete_food	Adds or removes food log entries
set_water, add_weight	Updates water and weight tracking

Endpoint / Action	Purpose
read_article, toggle_save_article	Tracks article reading and saved articles
hire_nutritionist, hire_yoga, cancel_hire	Manages expert booking activities
claim_reward, complete_challenge	Handles rewards and daily challenge completion
admin_login, admin_logout, admin_stats	Admin authentication and reporting

3.4. Database

MySQL is used as the database system. The database name is bitbite_db. The SQL setup file creates the required tables, sample user, sample admin, meal data, articles, nutritionists, yoga coaches, rewards, and challenges for a fresh XAMPP/phpMyAdmin setup.

Core Table	Purpose
users	Stores registered user identity and login credential hash.
profiles	Stores personal health profile, goals, body data, activity, diet preferences, budget, and wearable preference.
food_logs	Stores user food entries with food name, emoji, calories, meal type, and time.
weight_logs	Stores weight history and notes.
water_logs	Stores current water cup count for each user.
meal_plans	Stores planned meals with calories, macros, and cost.
shopping_list	Stores grocery items with quantity, price, and checked status.
sleep_logs	Stores sleep date, hours, quality, bedtime, and wake time.
notifications	Stores user notifications and read status.
achievements	Stores achievement badges and unlock status.
user_settings	Stores reminder and report preferences.
wearable_logs	Stores wearable-style health metrics such as steps, calories burned, heart rate, SpO2, sleep hours, and HRV.
Extra Feature Table	Purpose
admins	Stores admin login credentials.
nutritionists	Stores available nutritionist profiles.
yoga_coaches	Stores available yoga coach profiles.
nutritionist_hires	Stores user nutritionist booking requests and payment information.
yoga_coach_hires	Stores user yoga booking requests and payment information.
research_hub_articles	Stores article content for the Research Hub.
research_article_reads	Tracks article read count per user.
reward_catalog	Stores available reward items.
user_rewards	Stores reward claim status and delivery information.
daily_challenges	Stores daily challenge definitions.
user_challenge_completions	Tracks completed challenges and streak-related activity.
saved_articles	Stores user-saved Research Hub articles.

3.5. Testing

Testing was performed manually in XAMPP using the browser, phpMyAdmin, and a backend connection test file. The testing process focused on verifying frontend-to-backend communication, data insertion, data retrieval, and UI updates after database changes.

Test Area	Expected / Verified Result
Database Connection	Opened backend.php?action=me and confirmed JSON output. Not logged in response means backend and database connection are working.
Signup/Login	Created new accounts, verified confirm password validation, checked users and profiles tables.
Food Log	Added custom foods such as local food names and confirmed that the same

Test Area	Expected / Verified Result
	name appears on the frontend and in food_logs.
Water Log	Changed cup counts and verified updates in water_logs.
Weight Log	Logged new weight and verified insert in weight_logs and profile update.
Sleep Log	Added sleep hours and quality and verified database insertion.
Research Hub	Opened articles and tested read/save actions.
Bookings	Submitted nutritionist and yoga coach hire requests and verified booking tables.
Admin Panel	Logged in with admin credentials and reviewed admin statistics.

3.6. Git

Version control can be managed with Git by keeping frontend, backend, database, and documentation changes in separate commits. A recommended workflow is to use branches for frontend updates, backend/database updates, feature implementation, bug fixing, and final report/documentation. For final submission, the stable version should be tagged or stored in the main branch.

3.7. AI/ML Models (optional / if any)

The current BitBite implementation does not integrate an external AI/ML model or cloud AI API. The AI Coach is implemented as a rule-based assistant using JavaScript logic, user profile values, calorie/macro calculations, sleep/water data, and predefined insight messages. This approach provides an AI-like coaching experience for the project demo while keeping the system simple enough to run fully in XAMPP.

4. Future Work

- Integrate a real AI model or nutrition recommendation engine to generate fully dynamic meal plans and personalized coaching messages.
- Add real wearable device API integration for Google Fit, Fitbit, Apple Health, Garmin, or Samsung Health.
- Add secure online payment gateway support for nutritionist and yoga coach bookings.
- Build a mobile app version using Flutter or React Native.
- Add email verification, password reset, and stronger account security features.
- Add image-based food recognition so users can upload food photos for calorie estimation.
- Improve admin analytics with charts for active users, bookings, article engagement, and reward delivery status.
- Add role-based access control and audit logs for admin actions.
- Add notification reminders through email or push notifications.
- Deploy the project to a live web server with production database backup and SSL support.

5. Conclusion

BitBite successfully demonstrates a full-stack software engineering project for nutrition and wellness management. It includes user authentication, onboarding, health profile management, calorie and macro calculation, food logging, water and weight tracking, sleep logging, dashboard visualization, AI-style coaching, expert hiring, Research Hub activity, rewards, daily challenges, and admin reporting.

The project outcome is a working local web application that can be installed in a lab PC through XAMPP by placing the frontend/backend files in htdocs and importing the SQL database setup file in phpMyAdmin. Through this project, the team learned how to connect frontend JavaScript with PHP backend APIs,

manage sessions, design a MySQL schema, validate user input, store user activities, and organize a complete software system from requirements to implementation.

Appendix A: Demo Credentials and Setup Summary

Demo User Login: araf@example.com / 12345678

Demo Admin Login: admin@bitbite.com / admin12345

Local Setup Path: C:\xampp\htdocs\bitbite\

Local URL: <http://localhost/bitbite/index.html>

Database Name: bitbite_db

Setup Steps

8. Install and open XAMPP.
9. Start Apache and MySQL.
10. Create a folder named bitbite inside C:\xampp\htdocs\.
11. Copy index.html, style.css, script.js, backend.php, test_connection.php, and the SQL file into the project folder.
12. Open phpMyAdmin and run/import the database setup SQL file.
13. Open <http://localhost/bitbite/index.html> in a browser.
14. Use the demo credentials or create a new account to test the application.